

Johny Yong Jun Siang

Johnnyong10@gmail.com | +65 93726829 | <https://johnnyong.github.io/DeveloperPortfolio/> | www.linkedin.com/in/johnnyong

SUMMARY

Software Developer and B.Sc. (Hons) Interactive Media & Game Development student at Digipen/SIT with 5+ years of experience in **C++, C#, Python**, and **JavaScript**. Skilled in **game systems engineering**, **multiplayer networking**, **custom engine development**, and web technologies (**React**, **HTML**, **CSS**). Extensive **Unity** expertise across **VR/MR**, gameplay systems, shaders, and cross-platform deployment, with proven ability to lead and deliver projects from design to deployment.

SKILLS

Programming & Scripting: C, C++, C#, Python (Intermediate), Java (Intermediate), Kotlin (Intermediate), JavaScript, HTML, CSS, Lua

Networking & Backend: Photon Networking, Forge Networking, Firebase, Azure DevOps, REST-style data handling, JSON

Software & Tools: Git, GitHub Actions (CI/CD), Visual Studio 2022, Visual Studio Code, Node.js, React, Flutter, Android Studio, Unity, Unity 3D, Unreal Engine, Qt GUI, Anaconda, MkDocs, Figma, Canva, Photoshop

AI / ML Tools: Unity ML-Agents, PyTorch (training pipelines), Reinforcement Learning

Project Management & Design: Technical Design Documentation, Cross-functional Team Leadership, UI/UX Design, Agile Workflow

Spoken Languages: English (Fluent), Mandarin (Fluent)

NOTABLE PROJECTS

Developer Portfolio

December 2025 - December 2025

- Previously created my portfolio on Wix.com, I decided to rebuild my personal developer portfolio using **React**, with a focus on clean structure and reusable components.
- Hosted on GitHub Pages with a **GitHub Actions CI/CD pipeline** for automated build and deployment.
- Used **HTML**, **CSS**, and **JavaScript** to design and style the site, making sure it works well across different screen sizes.
- Set up the project with **Node.js tooling** for local development and builds.

Gravity

September 2024 - April 2025

- Led a 9-member team to develop a physics-driven top-down survival game built on a **custom C++ engine**, managing the product roadmap and technical milestones.
- Designed the engine's **Entity Component System (ECS)**, including a Game Object Factory and a Tag/Layer Manager to optimize object handling and performance.
- Authored comprehensive **Technical Design Documentation** using **MkDocs**, maintaining a live version via GitHub Pages for team alignment and version control.

Independent Reinforcement Learning Agent

January 2026 - January 2026

- Independently designed and trained a **reinforcement learning agent** using **Unity ML-Agents**, leveraging **PyTorch-based training pipelines**.
- Implemented observation spaces, action spaces, and reward shaping to study agent behavior and learning outcomes.
- Conducted iterative experimentation on training parameters and environments to evaluate policy convergence and performance.
- Gained hands-on exposure to **AI training workflows**, **inference behavior**, and **experimentation practices** in a real-time simulation environment.

EXPERIENCE

Mentor, Singapore Institute of Technology Community Challenge

Singapore Institute of Technology - Part Time

October 2025 - October 2025

- I volunteered to mentor teams in using **VittaScience AI** posture & image recognition tools for prototype development.
- Supported students across 5 days on their programming workflows, UI/UX design decisions, and technical troubleshooting.

Software Engineer

YJP Surveyors Pte Ltd - Internship

June 2020 - August 2020

- Contributed to a HoloLens mixed reality project built in Unity, enhancing visualisation of underground pipes and connections normally hidden by walls or flooring.
- Implemented **voice recognition** features to improve user interaction and accessibility.
- Gained hands-on experience with point cloud data and **LiDar technology** to support spatial mapping and alignment.
- Delivered a fully functional prototype that was commended for its effectiveness and innovation.

EDUCATION

B.Sc.(Hons), Interactive Media & Game Development

Singapore Institute of Technology / DigiPen Institute of Technology

August 2023 - April 2027

Diploma in Game Development and Technology

Nanyang Polytechnic

April 2018 - April 2021